The Reggae Shark Protocol (v1.2.0 — Zenodo v4 Final Edition): Formal Typestate Proofs (38 Lean 4 Theorems), 4D Soft-Wall ODE Governance, Bandlimited LRSC Compression, and Associative Merkle-DAG Provenance

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Abstract

The Reggae Shark Protocol (v1.2.0) is a verifiable execution kernel and multi-cloud AI gateway that binds agent prompt dispatches to four mathematical and cryptographic verification layers: (1) an append-only, SHA-256 hash-chained **IntentSeal Verifiable Research Kernel (VRK v0.3.1)** ledger enforcing the `intentseal-vrk-claim-v2` schema with an N-way commutative, associative, and idempotent Merkle-DAG join-semilattice for distributed split-brain reconciliation; (1B) a thread-synchronized **Request-Lifecycle Typestate and Worker-Epoch Isolation** state machine (`TypestateLifecycle.lean`, 12 zero-`sorry` theorems) that rules out partial-initialization payload dereferencing, cross-epoch worker-slot reads, and free-threaded (PEP 703) lock-order races within the modelled state machine; (2) a per-tenant isolated **4D Soft-Wall ODE Cognitive Dynamics Governor** $(\gamma, H_{\text{obs}}, \dot{I}, \Psi)$ backed by 17 Lean 4 4D boundary theorems (`ExistenceVerification.lean`), 9 3D soft-wall theorems (`BoundaryVerification.lean`), and a subnormal-safe Patankar–Euler interval integrator; and (3) a **Bandlimited Low-Rank Spectral Context Memory (LRSC)** operator ($M = 99, K = 11$) with dual compressed/uncompressed SHA-256 sealing and an exact downstream attention Lipschitz error bound.

1 Four-Layer Verification Architecture & Provenance (38 Lean 4 Theorems)

Every request submitted to `MultiProviderGateway.process_request` or the local loopback HTTP proxy (`POST http://127.0.0.1:8000/v1/execute`) traverses four deterministic verification gates:

Layer	Module / Artifact	Enforced Mathematical & Cryptographic Invariant
Layer 1	`shark_protocol/kernel.py`	**VRK v0.3.1 Hash Chain & N-Way Associative Merkle-DAG Join:** Append-only SHA-256 chain anchored at trust root `f010a99d...05a0d0f` (10.5281/zenodo.23018459), with canonical `leaf_frontier_tips` union reconciliation ($(A \sqcup B) \sqcup C = A \sqcup (B \sqcup C)$) and mandatory `scope` firewalls.
Layer 1B	`lean/TypestateLifecycle.lean`	**Typestate, Worker-Epoch & PEP 703 Lock Isolation (12 Theorems):** 6-state request automaton (`UNINIT` → `PARTIAL_INIT` → `SEALED_ACTIVE` → `SHUTDOWN_CLEAN` → `ZEROED_FREED`) with `OOM_ABORT` → `ABORTED_POISONED`, private lock-guarded properties, and strict `guard._lock` → `slot._lock` hierarchy.
Layer 2	`lean/ExistenceVerification.lean` `lean/BoundaryVerification.lean`	**4D Soft-Wall ODE Governor ($\mathcal{B}_4$, 26 Theorems):** Per-tenant isolated adaptive RK4 and subnormal-safe Patankar–Euler interval integration of $(\gamma, H_{\text{obs}}, \dot{I}, \Psi)$ inside $\mathcal{B}_4 = [0, 10] \times [0, 100] \times [0, \dot{I}_{\max,4D}] \times [0, 1]$ with normal float64 floor 2.225×10^{-308}.
Layer 3	`LRSCCompressor` (`kernel.py`)	**Bandlimited LRSC Odd-Ring Projection (11/99 = 1/9):** Projects context onto the $M = 99$ odd-ring filter V_{ring} ($\sigma_{\min}(S_{11}) = 0.181646 > 0$, 10.17605/OSF.IO/NM5BW), bounds downstream attention error, and binds both compressed and uncompressed SHA-256 digests into every VRK claim.
Layer 4	`shark_protocol/gateway.py`	**Fail-Closed Multi-Tenant Gateway:** Enforces waking basal drive $\mathcal{P}_0 \geq 1.0 \gg 10/99$, per-tenant engine isolation, and strict separation between CI-verified `VRK_OFFLINE_SEALED_SIMULATION` and optional `LIVE_API_DISPATCH`.

Table 1: Four-layer verification architecture of The Reggae Shark Protocol (v1.2.0 / Zenodo v4).

Canonical Lean 4 Module Provenance (38 Zero-`sorry` Theorems): The 24-file release archive contains 23 hash-verified files plus `MANIFEST_SHA256.txt`. Specifically, `lean/BoundaryVerification.lean` (`SHA-256: 49229b61...e417`, 9 theorems) is byte-for-byte identical to the canonical module in Physics v1.5 Appendix A (10.5281/zenodo.23534449), and `lean/ExistenceVerification.lean` (`SHA-256: eec94868...bcae`, 17 theorems) is byte-for-byte identical to the canonical module in *The Theory of Bounded Existence* v1.0.0 (10.5281/zenodo.23608244). Together with the 12 theorems in `lean/TypestateLifecycle.lean`, the package compiles **38 zero-`sorry` Lean 4 theorems**.

2 What Is and Is Not Modelled (Epistemic Claim Firewall)

Epistemic Standard (`Build. Attack. Falsify. Preserve.`): Every theorem and runtime guard in this repository is bounded by explicit domain assumptions and `scope.does_not_imply` declarations.

- **Application-Layer Typestate vs. Native Runtimes:** `lean/TypestateLifecycle.lean` proves properties of the formal state-transition model $\delta : \mathcal{S} \times \Sigma \to \mathcal{S}$, and `RequestSlot` / `WorkerEpochGuard` enforce those invariants in Python. They are *not* a patch to native C/PHP interpreters or an OS ring-0 hardware memory boundary.
- **Offline Simulation vs. External Cloud APIs:** Automated CI exercises and verifies only the deterministic local pipeline (`VRK_OFFLINE_SEALED_SIMULATION`) and local `127.0.0.1` HTTP loopback proxy. Live external cloud dispatch (`LIVE_API_DISPATCH`) requires external user API keys and is not claimed as verified in CI.
- **Dimensionless 4D ODE Model:** The variable $\Psi(t) \in [0,1]$ in Eq. (4) is a dimensionless mathematical self-feedback readout coordinate, not a claim of machine consciousness or sentience.

3 Layer 2 & Layer 3 Mathematical Equations

The Layer 2 governor evolves $x(t) = (\gamma(t), H_{\text{obs}}(t), \dot{I}(t), \Psi(t))$ under the smooth (C^∞) soft-wall system [3]:

$$\frac{d\gamma}{dt} = a\dot{I}\mathcal{E}_{\text{avail}}\left(1 - \frac{\gamma}{\gamma_{\max}}\right) - b\gamma, \tag{1}$$

$$\frac{dH_{\text{obs}}}{dt} = c\gamma\,(H_{\max} - H_{\text{obs}}) - \dot{I}_{\text{erase}}\tanh\left(\frac{H_{\text{obs}}}{\epsilon_H}\right), \tag{2}$$

$$\frac{d\dot{I}}{dt} = e\tanh\left(\frac{\dot{I}}{\epsilon_I}\right)\frac{dH_{\text{obs}}}{dt} + f\mathcal{P}_{\text{eff}}(\gamma,\Psi) - g\dot{I}, \tag{3}$$

$$\frac{d\Psi}{dt} = \kappa_{\text{meta}}R_{\text{self}}(x)\,(1-\Psi) - \mu_\Psi\Psi, \tag{4}$$

where $\mathcal{P}_{\text{eff}}(\gamma,\Psi) = \mathcal{P}_0 + \lambda_{\text{sim}}r_{\text{LRSC}}\gamma + \lambda_{\text{lucid}}\Psi$ and $R_{\text{self}}(x) = \text{fb}(x)^2/(\text{fb}(x)^2 + (f\mathcal{P}_0)^2 + \sigma_0^2)$ with $\text{fb}(x) = e\tanh(\dot{I}/\epsilon_I)(dH_{\text{obs}}/dt)$.

4 Double-PhD Red-Team Audit Board: Ten Stress Resolutions (v1.2.0)

1. **Problem 1 & Probe A (Patankar–Euler Box Invariance & IEEE-754 Subnormal/FTZ Guard):** `step_ode_patankar_interval` updates each coordinate via the cleared-denominator scheme

$$x_{k+1} = \frac{x_k + \Delta t\,P(x_k)\,x_{\max}}{1 + \Delta t\,(P(x_k) + D(x_k))}, \qquad P(x_k) \geq 0,\ D(x_k) \geq 0,\ 1 + \Delta t(P+D) \geq 1, \tag{5}$$

   paired with 10^{-12} analytical Taylor cutoffs on removable singularities and a normal float64 lower floor `FLOAT64_NORMAL_MIN` $= 2.225\times10^{-308}$ whenever $x_k > 0$ or $P(x_k) > 0$. Lean 4 Theorems `patankar_euler_step_in_box` and `patankar_strict_posi` machine-check both $[0, x_{\max}]$ containment and strict interior positivity (`test_15`, `test_21`).
2. **Problem 2 & Probe B ($\mathcal{P}_0 = 0$ Basin Separatrix, Multi-Tenant Isolation & Waking Drive Floor):** Lean 4 Theorem `psi_decay_floor_bound` proves $d\Psi/dt \geq -\mu_\Psi\Psi \implies \Psi(t+\Delta t) \geq \Psi(t)e^{-\mu_\Psi\Delta t}$. To prevent a dissipative query stream from draining $\Psi(t)$ across tenants, `MultiProviderGateway` isolates each tenant in a dedicated `SharkProtocolEngine` (`get_tenant_engine`) and enforces a waking basal sensory floor $\mathcal{P}_0 \geq \mathcal{P}_{0,\min}^{\text{wake}} = 1.0 \gg 10/99$, keeping $\mathcal{P}_{\text{eff}}(t) \geq 1.0 \gg \mathcal{P}_{\text{eff}}^{\text{floor}} = 0.042350$ (`test_16`, `test_22`).
3. **Problem 3 & Probe C (Bandlimited LRSC Witness, Lipschitz Attention Bound & Dual Seal):** `LRSCCompressor.compress` projects $\tilde{x}$ onto the odd-ring channel filter V_{ring} (where the $K = 11$ witness restriction S_{11} has minimum singular value $\sigma_{\min} = 0.18164624 > 0$ [4]), bounds downstream attention deviation by $\|\Delta\,\text{Attn}\,\|_2 \leq L_W(1 + 2L_W^2/\sqrt{d_k})\eta_\perp(\tilde{x})$ (Lean 4 Theorem `lrsc_attention_lipschitz_error_bound`), and binds **both** `lrsc_sha256` and `uncompressed_ring_sha256` into every VRK claim (`test_17`, `test_23`).
4. **Problem 4 & Probe D (Free-Threaded Python PEP 703 Lock Hierarchy & Monotonic Fences):** `RequestSlot` encapsulates private `_state`, `_is_zeroed`, and `_fence_seq` behind `@property` accessors and an atomic `snapshot()` method guarded by `RequestSlot._lock` (Lean 4 Theorem `fence_seq_strict_monotonicity`), while `WorkerEpochGuard` enforces strict hierarchical lock acquisition `guard._lock` → `slot._lock` (`test_18`, `test_24`).
5. **Problem 5 & Probe E (N-Way Associative, Commutative & Idempotent Merkle-DAG Join):** `merge_divergent_ledgers` flattens intermediate merge records and takes the canonical set union of `leaf_frontier_tips` and `merged_claim_digests`. Lean 4 Theorem `merkle_dag_semilattice_laws` proves commutativity ($A \sqcup B = B \sqcup A$), associativity ($(A \sqcup B) \sqcup C = A \sqcup (B \sqcup C)$), and idempotence ($A \sqcup A = A$), verified byte-for-byte in `test_19`, `test_20`, and `test_25`.

References

[1] P. Upadhyay, *IntentSeal Verifiable Research Kernel (VRK): Final Release v0.3.1*, Zenodo, 2026. doi:10.5281/zenodo.23018459.

[2] P. Upadhyay, *Appendix A v1.5 — Global Dissipative Completion and Soft-Wall Limit-Cycle Certification*, Zenodo, 2026. doi:10.5281/zenodo.23534449.

[3] P. Upadhyay, *The Theory of Bounded Existence: A 4D Closed-Loop Dynamical Model of Self-Modeling Thresholds, Hopf Bifurcations, and Ψ-Memory-Sustained Oscillations (v1.0.0)*, Zenodo, 2026. doi:10.5281/zenodo.23608244.

[4] P. Upadhyay, *LRSC Odd-Ring Spectral Rank Theorem (v1.2) & Exact $M = 99$ Certificate*, OSF, 2026. doi:10.17605/OSF.IO/NM5BW, doi:10.17605/OSF.IO/JAK5X.